Biodegradation (2026) 37:17
https://doi.org/10.1007/s10532-025-10239-w

ORIGINAL PAPER

Biosorption-based elimination of hexavalent chromium using *Bacillus cereus* F4810/72 isolated from tannery effluents

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Received: 26 October 2025 / Accepted: 13 December 2025
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Abstract The leather industry of West Bengal continuously discharges chromium-laden effluents, with hexavalent chromium [Cr(VI)] being particularly persistent and toxic. Hexavalent chromium released from tannery effluents poses serious environmental and public health hazards due to its high mobility, solubility, and carcinogenicity. Biosorption offers a sustainable and cost-effective alternative to conventional physicochemical remediation methods. In this study, the biosorption potential of *Bacillus cereus* F4810/72, a strain previously isolated from tannery wastewater, was systematically evaluated for Cr(VI) removal under varying physicochemical conditions. Batch experiments were conducted to investigate the influence of initial metal concentration (10–1000 µg/mL), contact time (10–100 min), pH (1–9), biomass concentration (1–4 g/L), temperature (10–60 °C), and agitation speed (10–120 rpm) on biosorption efficiency. Maximum Cr(VI) removal (approximately 79–80%) was achieved under optimized conditions: 250 µg/mL initial Cr(VI), 4 g/L biomass, pH 3, 50 °C, 70 min contact time, and 80 rpm agitation. Langmuir and Freundlich isotherm models were applied to elucidate adsorption behavior. Nonlinear Langmuir modeling indicated monolayer adsorption with a maximum capacity (Q_{max}) of 28.57 mg/g, whereas the Freundlich model ($R^2 = 0.98$) suggested adsorption onto a heterogeneous surface. Regeneration studies showed that 2 M HCl and a combined HNO_3–ascorbic acid eluent exhibited superior desorption efficiencies, maintaining high uptake–release performance across three cycles. Overall, *B. cereus* F4810/72 demonstrates strong Cr(VI) biosorption and regeneration

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10532-025-10239-w.

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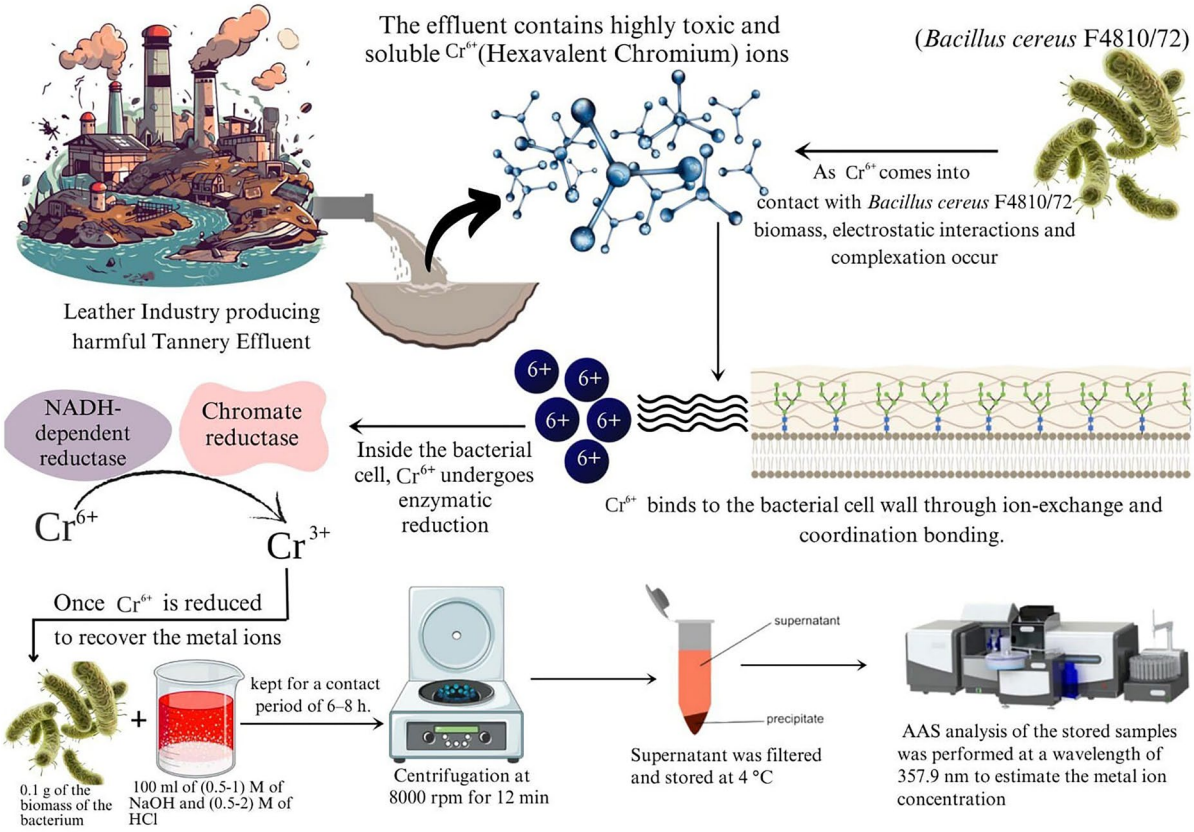
Published online: 23 December 2025

potential, highlighting its viability as a low-cost, eco-friendly biosorbent for chromium-contaminated industrial wastewater systems.

Graphical Abstract



Keywords *Bacillus cereus* · Chromium removal · Cr(VI) remediation · Tannery waste water · Bioremediation · Biomass · Biosorption

Introduction

Rapid industrialization and urban expansion have intensified the release of toxic contaminants into natural ecosystems, causing ecological imbalance, biodiversity loss, and bioaccumulation across trophic levels (Liang et al. 2019; Chormare et al. 2022). Although essential metal ions such as Mg^{2+}, Mn^{2+}, Ca^{2+}, Ni^{2+}, Na^{+}, and Zn^{2+} are required for cellular metabolism, non-essential metals—including chromium, mercury, and lead—are toxic even at trace concentrations and pose serious environmental and health risks (Balali-Mood et al. 2021). Tannery wastewater represents a major source of chromium contamination because chromium salts are widely used during leather tanning and are frequently discharged without adequate treatment (Arti and Mehra 2023). Chromium predominantly exists in two stable oxidation states: trivalent Cr(III) and hexavalent Cr(VI). Cr(III) is generally regarded as less toxic and participates in glucose and lipid metabolism, whereas Cr(VI) is highly soluble, mobile, and carcinogenic in nature (Swaroop et al. 2019; Monga et al. 2022). Exposure to Cr(VI)

is associated with oxidative stress, DNA damage, respiratory illnesses, malignancies, and detrimental effects on aquatic organisms and plants (Urbano et al. 2008; Shin et al. 2023). Structurally and chemically, Cr(VI) exhibits far greater toxicity than Cr(III) due to differences in membrane transport and redox reactivity. Cr(VI), present primarily as chromate (CrO_4^{2-}), mimics sulfate and phosphate, enabling efficient cellular entry through anion transporters. Once inside the cell, Cr(VI) undergoes stepwise reduction to Cr(V), Cr(IV), and finally Cr(III), generating reactive oxygen species (ROS) that drive oxidative stress and genotoxicity. The intracellular Cr(III) produced during this reduction further forms DNA–protein crosslinks and Cr–DNA adducts. In contrast, Cr(III) exists as large, hydrated cations with poor membrane permeability, limited redox activity, and minimal interaction with intracellular macromolecules. The high membrane permeability and strong oxidizing nature of Cr(VI) account for its pronounced cytotoxic, mutagenic, and carcinogenic effects, whereas Cr(III) remains comparatively benign under similar exposure conditions (Bagchi et al. 2001; DesMarais and Costa 2019; Krawic and Zhitkovich 2023). These concerns highlight the need for effective and sustainable remediation strategies for Cr(VI)-contaminated effluents. Conventional treatment methods—chemical precipitation, ion exchange, membrane filtration, electrochemical treatment, and solvent extraction—although effective, are limited by high operational costs, complex maintenance, and the generation of large volumes of toxic sludge (Gunatilake 2015; Yadav et al. 2021; Qasem et al. 2021). These limitations have driven the advancement of cost-effective and environmentally benign technologies. Biosorption has gained prominence as a promising alternative due to its operational simplicity, low cost, high metal-binding efficiency, and negligible secondary sludge generation (Yaashikaa et al. 2024; Abdi and Kazemi 2015). Low-cost biosorbents derived from agricultural residues—such as maize biomass, corn stalks, and groundnut shells—have shown high sorption capacities for heavy metals including Pb(II), Hg(II), and Cr(VI) (Bayuo et al. 2025b; Bayuo et al. 2022; Bayuo et al. 2025a; Bayuo et al. 2024a; Bayuo et al. 2024b; Bayuo et al. 2023; Bayuo et al. 2019). In another study by the same group, the groundnut husk biosorbent demonstrated effective reusability for up to five regeneration cycles, retaining removal efficiencies of 53.5% for Cr(VI) and 54.6% for Pb(II) after the fifth cycle (Bayuo et al. 2020). Biochar-based systems further enhance sorption performance owing to their high surface area, surface functional groups, and ability to immobilize microbial cells or enzymes, which improves degradation and adsorption efficiency (Imam et al. 2021; Souza Junior et al. 2020). Sustainable biological treatment strategies such as microbial fuel cells and microbial consortia have also demonstrated substantial potential in pollutant removal and water quality improvement, underscoring their relevance in modern wastewater treatment (Dixit et al. 2025; Velmurugan and Pandian 2024). Microbial biosorbents, particularly bacteria, are extensively studied for metal adsorption due to the presence of active functional groups (carboxyl, phosphate, amino, and hydroxyl groups) on their cell walls, enabling ion exchange, complexation, and micro-precipitation mechanisms (Vijayaraghavan and Yun 2008; Aryal and Liakopoulou-Kyriakides 2015; Cai et al. 2009). Biosorption performance is strongly influenced by environmental parameters including pH, temperature, biomass concentration, and contact time (Aryal and Liakopoulou-Kyriakides 2015). Bacillus species are especially efficient biosorbents owing to their tolerance to metal stress, rapid growth, and abundant surface binding sites. Previous studies have shown that *Bacillus cereus* F4810/72—isolated from tannery wastewater—exhibits substantial Cr(VI) removal capacity and produces industrially relevant extracellular enzymes such as protease, lipase, and keratinase (Bhattacharjee et al. 2021). Given the increasing industrial interest in biosorption technologies, optimizing operational parameters is essential to maximize metal uptake and ensure scalability (Pham et al. 2022). Therefore, the present study evaluates the biosorption performance of *Bacillus cereus* F4810/72 biomass for Cr(VI) removal from tannery effluents, with a focus on optimizing pH, temperature, contact time, and biomass dosage. The novelty of this work lies in the multifaceted functional role of *B. cereus* F4810/72: the strain not only exhibits strong biosorptive and bioaccumulative capacities for chromium but also transforms chromium from a dynamic bioavailable pool into a more stable reservoir, while concurrently producing industrially significant enzymes. Furthermore, equilibrium and kinetic modeling approaches are employed to elucidate the underlying adsorption mechanisms. Overall, the findings aim to

contribute to the development of a robust, economically viable, and environmentally sustainable strategy for chromium remediation in industrial wastewater systems.

Previous study

In our earlier investigation, *Bacillus cereus* F4810/72 was identified as an efficient chromium-reducing bacterium capable of transforming hexavalent chromium (Cr^{6+}) into the less toxic trivalent form (Cr^{3+}) (Bhattacharjee et al. 2021). Water samples collected from Bantala Leather Complex in West Bengal for microbial screening were subjected to serial dilution (10^{-1}–10^{-4}), and 0.1 mL from each dilution was spread onto nutrient agar plates supplemented with graded concentrations of chromium (1 µg/mL to 1 mg/mL). Plates were incubated at 37 °C for 24 h, after which morphologically distinct colonies were isolated and repeatedly sub-cultured to obtain pure strains (Bhattacharjee et al. 2021). Among the isolates, *B. cereus* F4810/72 exhibited the highest tolerance to chromium and demonstrated the ability to degrade leather-derived waste, indicating strong bioremediation potential. Based on these characteristics, this strain was selected for biomass preparation and subsequently employed in chromium biosorption experiments.

Experiments on biosorption

To optimize chromium biosorption by *Bacillus cereus* F4810/72, a series of batch experiments were conducted by systematically varying the operational parameters, including initial Cr(VI) concentrations (10–1000 µg/mL), contact times (10–100 min), pH values (1–9), biomass concentrations (1–4 g/L), temperatures (10–60 °C), and agitation speeds (10–120 rpm). For each experiment, the required amount of bacterial biomass was suspended in 100 mL of Cr(VI) solution of the designated concentration and incubated in 500 mL Erlenmeyer flasks on a rotary shaker under controlled temperature, pH, and agitation conditions. After incubation, the suspensions were centrifuged at 7000 rpm for 10 min to separate the biomass from the solution, and the residual Cr(VI) concentration in the supernatant was determined using an atomic absorption spectrophotometer (Perkin Elmer Analyst 200) at 357.9 nm. All biosorption experiments were performed in triplicate, and the results were expressed as mean ± standard deviation (SD) (Kalin et al. 2005).

The percentage of metal removal (biosorption efficiency) was calculated using the standard equation:

$$\text{Biosorption efficiency}(\%) = \frac{C_i - Cf}{C_i} \times 100$$

C_i is the initial metal ion concentration and C_f is the final concentration after biosorption (Shafiq and Alazba 2021).

Investigation of biosorption isotherms

Adsorption isotherms describe the interaction between the adsorbate and the surface of the adsorbent at equilibrium, providing insight into the capacity and mechanism of the adsorption process (Huang and Liu 2013; Liu and Liu 2008). The Langmuir model assumes monolayer adsorption onto a homogeneous surface with a finite number of identical adsorption sites. The non–linear form of the Langmuir equation is:

$$q_e = \frac{Q_{max} K_L C_e}{1 + K_L c_e}$$

Where Q_{max}(mg/g) is the maximum adsorption capacity and K_L (L/mg) is the Langmuir equilibrium constant related to the affinity of binding sites. C_e and q_e represent equilibrium concentration and equilibrium adsorption capacity respectively.

Linearized Langmuir analysis was also performed using the following form:

$$\frac{C_e}{q_e} = \frac{1}{Q_{max} K_L} + \frac{C_e}{Q_{max}}$$

A plot of C_e/q_e vs C_e was used to estimate q_{max} and K_L. To validate model fitness, non-linear regression was performed using iterative curve fitting, and statistical indicators—including coefficient of determination (R^2), sum of squared errors (SSE), average relative error (ARE), sum of absolute errors (SAE) were calculated.

The Freundlich isotherm describes adsorption on heterogeneous surfaces and assumes multilayer adsorption with sites of varying energies. The non-linear form of the Freundlich equation is:

$$q_e = K_F C_e^{\frac{1}{n}}$$

where K_F represents the adsorption capacity and 1/n describes adsorption intensity or surface heterogeneity.

The linear form was applied using the logarithmic transformation:

$$\ln q_e = \ln K_F + \frac{1}{n}\ln C_e$$

A plot of ln q_e vs ln C_e provides estimates of K_F and 1/n. The favorability of adsorption is indicated when $1/n < 1$, suggesting normal adsorption, whereas $1/n > 1$ indicates cooperative adsorption. Non-linear fitting accuracy was evaluated using R^2 and error functions to compare performance with the Langmuir model.

Desorption efficiency analysis

To evaluate the reusability of the biosorbent and the efficiency of metal ion recovery, desorption experiments were performed using chemical eluents. For each assay, 0.1 g of *Bacillus cereus* F4810/72 biomass was treated with 100 mL of (0.5–2) M HCl or (0.5–1) M NaOH and incubated for 6–8 h. After incubation, the mixtures were centrifuged at 8000 rpm for 12 min to separate the biomass from the eluent, and the resulting supernatants were filtered and stored at 4 °C for further analysis. Chromium concentrations in the recovered eluates were quantified using atomic absorption spectrophotometry at 357.9 nm (Ouyang et al. 2019), and the percentage of metal recovery was calculated using the standard formula described in previous studies (Shafiq and Alazba 2021).

Elution efficiency and regeneration potential of biosorbent

To evaluate the desorption efficiency and reusability of the biosorbent, chromium-loaded *Bacillus cereus* F4810/72 biomass was treated with 100 mL of different chemical eluants—HCl, HNO_3, ascorbic acid, EDTA, and NaOH—at concentrations ranging from 0.05 to 0.2 M. The desorption process was carried out at a controlled flow rate of 1 mL/min to ensure uniform exposure of the biomass to the eluants. Following each desorption cycle, the biomass was rinsed twice with 500 mL of deionized water to remove residual eluent and enable effective regeneration. The regenerated biomass was then reused in subsequent chromium biosorption cycles to assess its metal uptake efficiency, structural stability, and overall reusability across repeated operational cycles (Chandra Sekhar et al. 2004).

Statistical study

All experimental procedures were performed in triplicate ($n = 3$), and data are expressed as mean ± standard deviation (SD). Statistical significance was evaluated using two-way analysis of variance (ANOVA) in GraphPad Prism (version 5). Pearson's correlation coefficient was calculated to assess the strength of relationships between experimental variables. A p-value < 0.0001 was considered statistically significant. All graphs and data visualizations were generated using GraphPad Prism (version 5).

Results

Initial metal ion concentration plays a crucial role in determining the biosorption capacity of the biosorbent

The influence of varying initial concentrations of chromium ions on the biosorption efficiency of *Bacillus cereus* F4810/72 biomass was investigated within the range of 10 to 1000 µg/mL (Fig. 1). A notable enhancement in biosorption efficiency was observed as the metal ion concentration increased from 10 to 250 µg/mL, resulting in an improvement from 18.46 to 79.8% (Fig. 1). This trend indicates an increased interaction between metal ions and available binding sites on the biomass surface at lower concentrations. However, further elevation of the metal concentration beyond 250 µg/mL led to a decline in biosorption efficiency, which dropped to nearly 65% (Fig. 1). This reduction can be attributed to the saturation of active binding sites on the biosorbent surface, limiting the availability of free sites for additional metal ion adsorption and thus demonstrating a concentration-dependent saturation effect.

Effect of contact time on metal biosorption

The effect of contact time on chromium biosorption was investigated over a range of 10–100 min

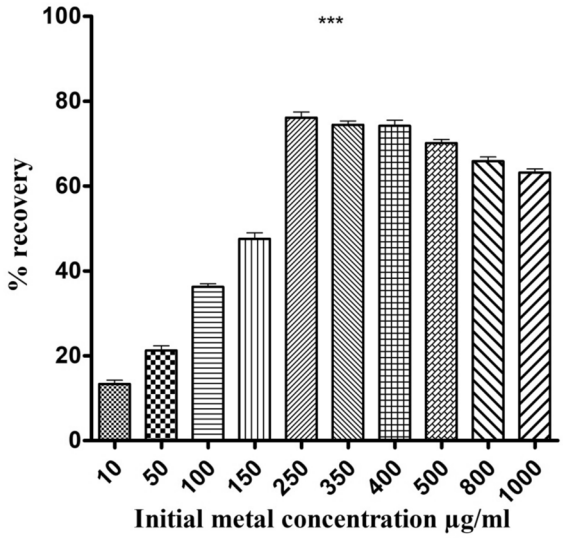


Fig. 1 Influence of initial metal ion concentration on chromium biosorption efficiency. The figure shows how varying initial concentrations of Cr(VI) (10–1000 µg/mL) influence the percentage of metal recovered by the biosorbent. Recovery (%) increases with rising metal load, reaching its maximum at 250 µg/mL, after which a gradual decline is observed, indicating saturation of available binding sites at higher concentrations. Error bars represent the standard deviation of triplicate experiments. Data is represented as ± SD. ***$p < 0.0001$ is considered as statistically significant

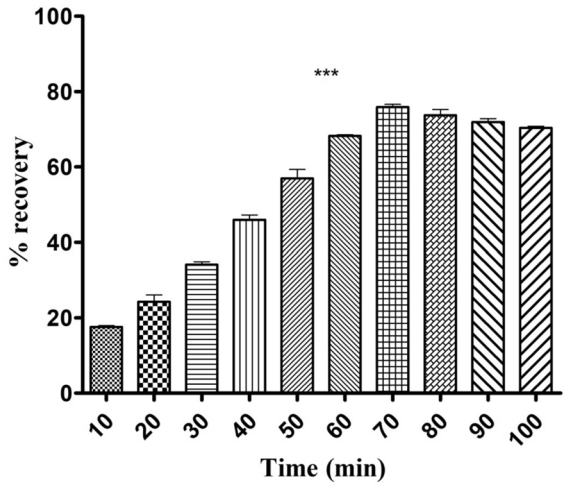


Fig. 2 Metal biosorption response over different contact times. Biosorption efficiency increased rapidly at initial stages and reached equilibrium after prolonged contact, indicating saturation of binding sites. Data represents mean ± SD of triplicate experiments. ***$p < 0.0001$ is considered as statistically significant

(Fig. 2). Biosorption efficiency increased progressively with time and reached a maximum of approximately 78% at 70 min. Beyond this point, no further appreciable increase in metal uptake was observed, indicating that the system had reached equilibrium. This plateau suggests that the majority of active binding sites on the biomass became saturated within the first 70 min, and extending the contact period did not significantly enhance chromium removal.

Biosorption capacity varies with changes in biomass concentration

The biosorption efficiency of *Bacillus cereus* F4810/72 was assessed at different biomass concentrations ranging from 1 to 4 g/L while keeping all other experimental conditions constant. A clear positive relationship was observed between biomass concentration and chromium uptake, with biosorption efficiency increasing progressively as the biomass dose increased from 1 to 4 g/L (Fig. 3). This improvement is likely due to the greater availability of functional surface binding sites at higher biomass concentrations. However, increasing the biomass concentration beyond 4 g/L did not

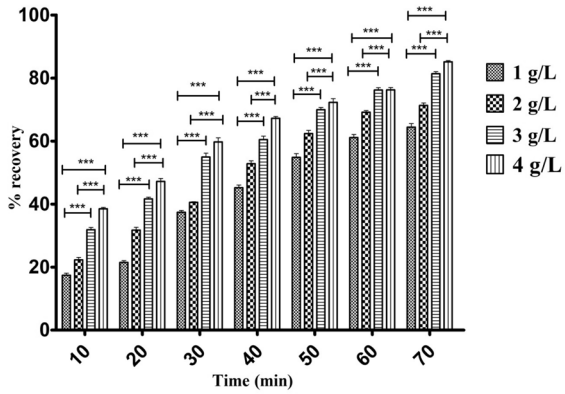


Fig. 3 Impact of varying biomass concentrations on metal ion uptake during biosorption. The figure illustrates the time-dependent recovery of Cr(VI) at different biomass dosages (1–4 g/L). For all dosages, recovery percentage increased progressively with contact time from 10 to 70 min, reflecting rapid initial metal binding followed by gradual attainment of equilibrium. Higher biomass concentrations (3 and 4 g/L) consistently showed greater recovery compared to lower dosages, indicating enhanced availability of active binding sites. Error bars represent the standard deviation. ***,$p < 0.0001$ is considered as statistically significant

produce any substantial enhancement in biosorption efficiency (data not shown). This plateau effect suggests that the system had reached saturation, where the available metal ions in solution were insufficient to occupy the additional binding sites, or that biomass aggregation at higher concentrations reduced the effective surface area available for adsorption.

Effect of pH on metal biosorption

The influence of pH on Cr(VI) biosorption by *Bacillus cereus* F4810/72 was examined across a pH range of 1–9. Biosorption efficiency varied markedly with pH, showing a substantial increase between pH 2 and 3. The maximum removal efficiency (~80%) was achieved at pH 3 after 70 min of contact (Fig. 4). Beyond this optimal pH, a gradual decrease in Cr(VI) uptake was observed. At pH 3 specifically, chromium removal ranged from about 35% to 78%, depending on the contact time. These results indicate that acidic conditions, particularly around pH 3, favor Cr(VI) biosorption by enhancing protonation of functional groups on the bacterial cell surface, thereby improving electrostatic attraction and metal-binding interactions.

Temperature dependent metal biosorption

The influence of temperature on the Cr(VI) biosorption capacity of *Bacillus cereus* F4810/72 was evaluated over a range of 10 °C to 60 °C. As shown in Fig. 5, increasing the temperature from 10 to 50 °C led to a marked improvement in biosorption efficiency, with a maximum uptake of approximately 78% observed at 50 °C. However, a decline in removal efficiency to nearly 62% was recorded at 60 °C, suggesting that temperatures above the optimal level may adversely affect the structural integrity of binding sites or the stability of functional groups involved in metal uptake.

Impact of agitation rate on metal ion uptake

The effect of agitation on the biosorption efficiency of *Bacillus cereus* F4810/72 was evaluated at shaking speeds ranging from 10 to 120 rpm. As shown in Fig. 6, chromium uptake increased progressively with agitation

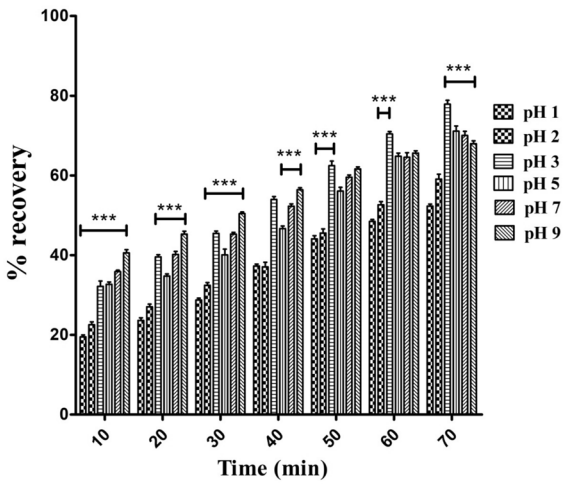


Fig. 4 Effect of pH on metal ion uptake during biosorption. The figure shows the percentage recovery of Cr(VI) at pH values ranging from 1 to 9 over contact times of 10–70 min. Recovery increased progressively with time for all pH conditions. Among the tested pH levels, the highest metal recovery was observed at pH 3, indicating that moderately acidic conditions favour chromium uptake. Error bars represent the standard deviation. Statistical significance between selected pH treatments is shown (***$p < 0.001$)

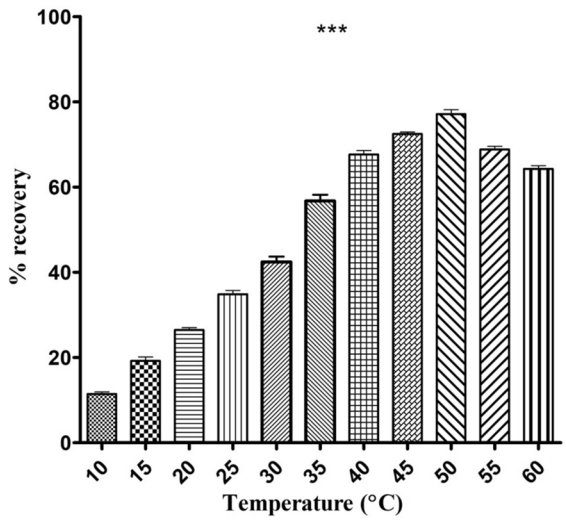


Fig. 5 Impact of varying temperatures on metal ion uptake during biosorption. The figure illustrates the percentage recovery of Cr(VI) at temperatures ranging from 10 to 60 °C. Metal recovery increased steadily with rising temperature, reaching its maximum at 50 °C, indicating enhanced biosorption activity at elevated temperatures due to increased molecular mobility and improved interaction between metal ions and binding sites. Data represents mean ± SD of triplicate experiments. ***$p < 0.0001$ is considered as statistically significant

rate and reached a maximum biosorption efficiency of approximately 80% at 80 rpm. Beyond this threshold, a decrease in removal efficiency was observed, likely due to the disruption of biomass–metal interactions or reduced contact time resulting from excessive turbulence. Thus, an agitation speed of 80 rpm was identified as the optimal condition for maximizing Cr(VI) biosorption.

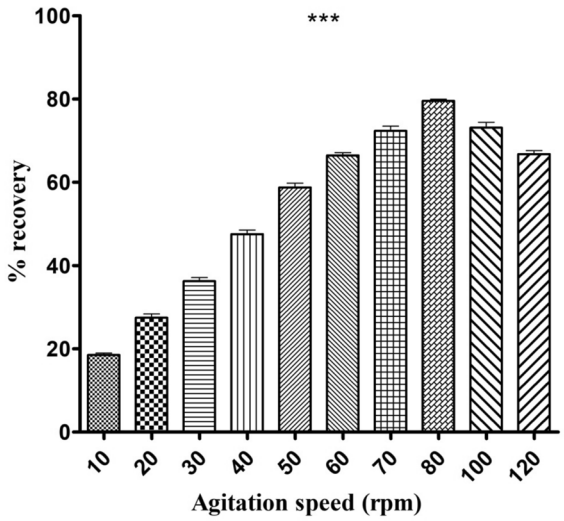


Fig. 6 Impact of varying agitation rates on metal biosorption capacity. The figure shows the percentage recovery of Cr(VI) at agitation speeds ranging from 10 to 120 rpm. Recovery increased steadily with higher agitation rates, reaching a maximum at 80 rpm, indicating improved mass transfer and enhanced contact between metal ions and biosorbent at optimal mixing. Data represents mean ± SD of triplicate experiments. ***$p < 0.0001$ is considered as statistically significant

Analysis of adsorption isotherm models

To understand the nature and efficiency of chromium biosorption by *Bacillus cereus* F4810/72, two widely accepted isotherm models—Langmuir and Freundlich—were applied. The equilibrium biosorption data were evaluated using Langmuir and Freundlich isotherm models in both linearized and nonlinear forms (Table 1).

- Langmuir isotherm: The linearized model yielded a Q_{max} of 114.40 mg g^{-1} and $K_L = 0.055$ L mg^{-1} with $R^2 = 0.985$, whereas the nonlinear fit gave $Q_{max} = 28.57$ mg g^{-1} and $K_L = 0.10$ L mg^{-1} with $R^2 = 0.986$. Error analysis indicated that the nonlinear model had significantly lower SSE (17.3 vs. 67.17), RMSE (1.70 vs. 3.34), ARE (12.8% vs. 29.6%), and χ^2 (1.43 vs. 8.25), suggesting a better fit.
- Freundlich isotherm: The linearized model gave $K_f = 2.07$ and $n = 1.44$ (1/n = 0.69, $R^2 = 0.985$) with high SSE (2302.5) and RMSE (19.6). The nonlinear Freundlich model resulted in $K_f = 3.53$, $n = 1.38$ (1/n = 0.598, $R^2 = 0.982$), with improved but still higher error values (SSE = 365.8, RMSE = 7.81) compared to nonlinear Langmuir.

Evaluation of desorption efficiency using bacterial biomass

The regeneration capacity of the *Bacillus cereus* F4810/72 biosorbent was assessed using different eluants. Among the tested agents, 2 M hydrochloric acid (HCl) exhibited the highest desorption efficiency, recovering approximately 80% of the bound chromium within 70 min (Fig. 7). In comparison, 1 M

Table 1 Langmuir and Freundlich isotherm parameters (linear and nonlinear) with error-function analysis

Model (method)	Parameters (reported)	R^2	SSE	ARE (%)	RMSE	χ^2
Langmuir (linearized)	$Q_{max} = 114.40$ mg·g⁻; $K_L = 0.055$ L·mg^{-1}	0.985	67.17	29.6	3.34	8.25
Langmuir (nonlinear)	$Q_{max} = 28.568$ mg·g^{-1}; $K_L = 0.10$ L·mg^{-1}	0.986	17.3	12.8	1.70	1.43
Freundlich (linearized)	$K_f = 2.07$; $n = 1.44$ 1/n = 0.69	0.985	2302.5	18.5	19.6	24.1
Freundlich (nonlinear)	$K_f = 3.53$; $n = 1.38$ 1/n = 0.598	0.982	365.8	25.0	7.81	7.73

SSE Sum of squared errors. *ARE* Average relative error (mean of |(exp − pred)/exp|× 100) *RMSE* Root mean squared error. χ^2 = chi-square ($\sum$(exp − pred)^2/pred)

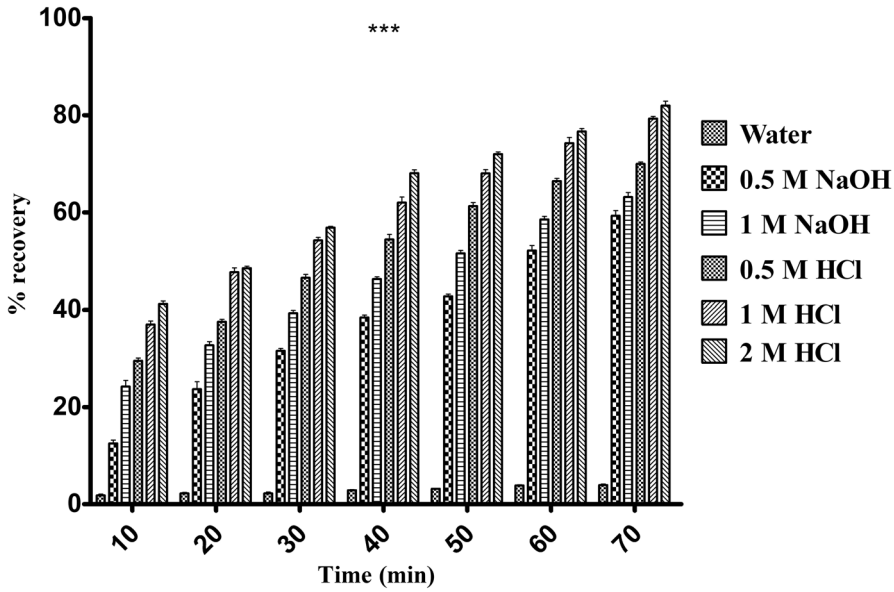


Fig. 7 Desorption study by the bacterial biomass. Desorption efficiency was evaluated over a contact period of 10–70 min using distilled water, NaOH (0.5 M and 1 M), and HCl (0.5 M, 1 M, and 2 M). Maximum desorption was observed using 2 M HCl at 70 min. Data represents mean ±SD of triplicate experiments. ***$p < 0.0001$ is considered as statistically significant

NaOH and 0.5 M HCl achieved lower recovery rates of about 65% and 70%, respectively. Treatments with deionized water and 0.5 M NaOH showed minimal desorption, indicating their limited ability to release adsorbed metal ions. These findings highlight 2 M HCl as the most effective eluant for regenerating the biosorbent.

Elution & regeneration study

Efficient desorption of metal ions from the biosorbent is essential to maintain its performance during repeated adsorption–desorption cycles. In the present study, elution experiments revealed that a combined solution of nitric acid and ascorbic acid (0.2 M) was the most effective eluant for chromium recovery, exhibiting higher desorption efficiency than hydrochloric acid (HCl) and EDTA (Fig. 8). When applied individually, nitric acid and ascorbic acid showed comparatively limited desorption performance. Regeneration experiments further demonstrated that the mixed nitric acid–ascorbic acid solution enabled consistent and effective chromium elution over three consecutive cycles (Table 3). These findings suggest that the biosorbent can be efficiently regenerated and reused multiple times; highlighting it's potential as a sustainable and cost-effective material for heavy metal remediation in aqueous systems.

Discussion

This study explores the potential of *Bacillus cereus* F4810/72 as a biosorbent for the effective removal of hazardous chromium (Cr) ions from tannery effluents. Chromium, particularly in its hexavalent form Cr(VI), is a highly toxic environmental pollutant. It is commonly released into surrounding ecosystems through wastewater discharge from the leather industry, leading to serious ecological and health concerns, especially in regions such as West Bengal (Maurya et al. 2023; Nigam et al. 2015; Tadesse et al. 2017). The biosorption process employed here offers a sustainable and eco-friendly alternative for heavy metal remediation. It relies on biological materials—either living or inactivated—to remove metal ions from aqueous environments (Kelany et al. 2023; Murthy et al. 2023). Compared to conventional chemical or physical treatment methods, biosorption is advantageous due to its cost-effectiveness, environmental compatibility, and efficiency (Yaashikaa et al. 2024; Rashid et al. 2021). The inherent metal-binding capabilities of microbial biomass, as demonstrated by *B. cereus* F4810/72, make it a promising agent for chromium removal. One key factor influencing biosorption efficiency is the initial concentration of chromium in the solution. The study establishes a clear positive relationship between metal ion concentration and biosorption capacity, attributed to the enhanced

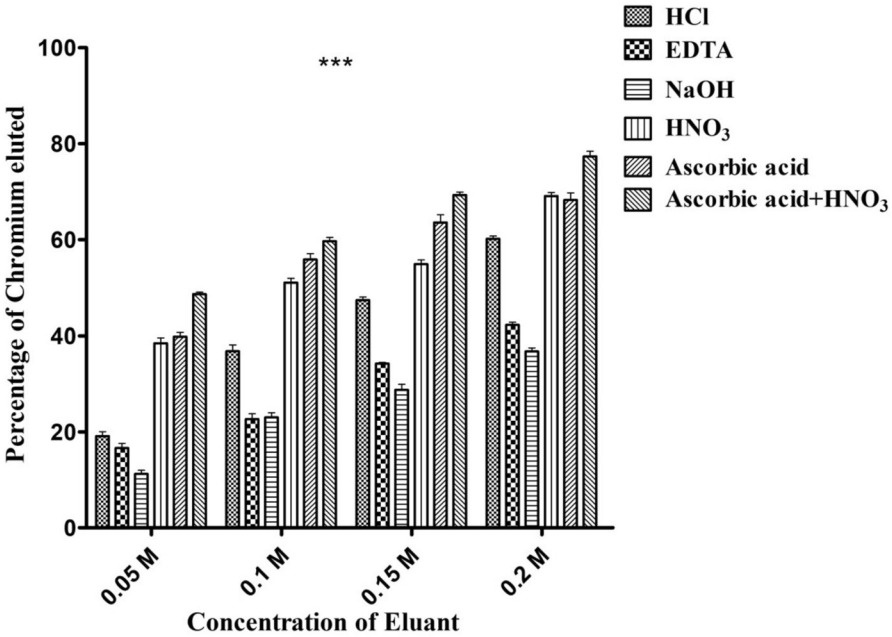


Fig. 8 Study on the elution efficiency of metal ions from treated biomass. The figure illustrates the percentage of chromium eluted from treated biomass using six eluents—HCl, EDTA, NaOH, HNO_3, ascorbic acid, and a combined ascorbic acid + HNO_3 system—tested at concentrations ranging from 0.05 to 0.2 M. Elution efficiency increased with rising eluent concentration for all treatments. Among the eluents, **a**scorbic acid + HNO_3 exhibited the highest desorption efficiency, followed by HNO_3 and ascorbic acid alone, indicating the strong ability of acidic and reducing environments to disrupt metal–biomass interactions. EDTA and NaOH showed comparatively lower desorption performance. Data represents mean ± SD of triplicate experiments. Statistical significance is indicated (***$p < 0.0001$)

driving force for metal binding at higher concentrations. As metal ion availability increases, more binding interactions occur at the biomass surface, resulting in improved biosorption, which rose from 16.46% at 10 µg/mL to nearly 79.8% at 250 µg/mL. At lower concentrations (10–250 µg/mL), a progressive increase in biosorption efficiency was recorded, rising from 19.46% to 79.8%. This trend suggests that the biosorbent surface possessed a high abundance of unoccupied functional groups—such as carboxyl, amino, phosphate, and hydroxyl moieties—that actively interacted with the available Cr(VI) ions. Under these conditions, the metal ion population remained lower than the total number of binding sites, enabling efficient mass transfer and rapid surface complexation. However, once saturation of the binding sites is reached, a decline in efficiency is observed nearly 62% as demonstrated in Fig. 1. At higher metal loadings, the fixed number of active sites on the bacterial cell wall becomes insufficient to accommodate the excessive Cr(VI) ions, resulting in competitive interactions and reduced adsorption efficiency. Additionally, elevated metal concentrations may induce electrostatic repulsion between accumulated metal ions on the biomass surface or cause partial inhibition of cell wall functional groups, further impeding metal–biomass interactions. This aligns with findings from Todorova et al., who reported reduced biosorption efficiency for lead, cadmium, and mercury by *Bacillus cereus* at elevated metal concentrations (Todorova et al. 2019). *Bacillus thurigenensis* has been reported to remove chromium ions from aqueous solution at a metal ion concentration of 250 mg/L (Şahin and Öztürk 2005). A study by Ansari et al., showed that biosorption of cadmium and nickel increased drastically at a metal concentration ranging between 50–400 µg/ml (Ansari and Malik 2007). These observations underscore the importance of optimizing metal ion concentrations in biosorption systems to prevent site saturation and ensure maximum removal efficiency. The impact of varying contact time on chromium biosorption was assessed

across a series of time intervals ranging from 10 to 100 min (Fig. 2). The results revealed that the maximum biosorption efficiency, approximately 78%, was achieved at 70 min of contact. Beyond this duration, no significant increase in biosorption was observed, indicating that equilibrium had been attained which suggests that most of the available binding sites on the biomass were saturated within the first 70 min, and further extension of the contact time did not notably enhance metal uptake. This behavior is characteristic of biosorption kinetics, where a rapid initial phase of metal uptake is followed by a slower approach to equilibrium. The initial rapid phase can be attributed to the abundance of available active binding sites on the surface of the biomass. At the beginning of the contact period, chromium ions in the solution encounter numerous vacant functional groups (e.g., carboxyl, hydroxyl, amino, phosphate) on the microbial cell wall that facilitate binding through mechanisms such as electrostatic attraction, ion exchange, coordination, and complexation. As the contact time increases, more chromium ions occupy these binding sites, leading to a gradual saturation of the biosorbent surface. The lack of significant increase in metal uptake beyond 70 min suggests that prolonged contact does not enhance biosorption, as desorption and adsorption rates become balanced. In a study using *Lactobacillus brevis*, the optimal concentration for the removal of lead was observed during a contact time of 12 h (Dai et al. 2019). However, the optimum contact time for the removal of Zn^{2+} was 30 min by *Pseudomonas aeruginosa* and *Bacillus cereus* (Joo et al. 2010). However when *Streptomyces rimosus* was used for the removal of zinc, the contact time increased to 4 h to achieve highest biosorption (Mameri et al. 1999). Determining the optimal contact time is crucial for the effective design of biosorption systems. This knowledge is especially important for large-scale industrial wastewater treatment, where fast and efficient processes are necessary for practical application. The biosorption efficiency of *Bacillus cereus* F4810/72 was evaluated at varying biomass concentrations ranging from 1 to 4 g/L, while maintaining all other experimental parameters constant. The findings demonstrated a positive correlation between biomass concentration and metal uptake, with increased biosorption efficiency observed as the biomass dose increased from 1 to 4 g/L (Fig. 3). However, further increases in biomass beyond 4 g/L did not result in any significant improvement in biosorption efficiency (data not shown), suggesting that a threshold level had been reached where all available metal ions had already been adsorbed, or particle aggregation may have reduced the effective surface area. This trend, consistent with observations in *Bacillus amyloliquefaciens* (Ramachandran et al. 2022) and *Staphylococcus xylosus* (Aryal et al. 2010), can be attributed to the increased number of active binding sites. Some studies even report decreased metal uptake at excessive biomass levels due to particle aggregation or limited diffusion (Aryal et al. 2012). In a study by Vishan et al., highest recovery of cadmium metal ions was achieved at biomass concentration of 2 g/L using *Bacillus badius* AK (Vishan et al. 2019). Similarly, at an optimum concentration of 1 g/L biosorbent dose maximum uptake of lead ions was observed using a marine bacterium *Bacillus xiamenensis* (Mohapatra et al. 2019). Optimization of biomass concentration is therefore crucial for balancing efficiency, cost, and resource use. The solution pH also plays a pivotal role in determining biosorption performance. Maximum chromium removal (~80%) occurred at pH 3.0 during an contact time of 70 min (Fig. 4). This is likely due to the chemical speciation of Cr(VI) in aqueous environments and the surface charge dynamics of the bacterial biomass. Under acidic conditions (pH 1–2), Cr(VI) predominantly exists as $HCrO_4^-$, a form that is more readily adsorbed due to electrostatic attraction to positively charged functional groups on the biomass surface. The low pH protonates functional groups such as OH, –COOH, and $–NH_2$, increasing the net positive charge on the bacterial cell surface and thereby enhancing attraction to anionic chromate species (Zha et al. 2024). In contrast, at higher pH values, deprotonation of these functional groups results in a negatively charged biomass surface, leading to electrostatic repulsion between the chromate anions and the biosorbent (Ayele and Godeto 2021). Our results align with prior findings involving *Mycobacterium* sp. and *Staphylococcus xylosus*, where maximal biosorption occurred at highly acidic pH (Aryal and Liakopoulou-Kyriakides 2014; Ziagova et al. 2007). However, at higher pH values, decreased biosorption is typically observed due to reduced protonation and potential metal precipitation, especially as Cr(III) hydroxide (Tan et al. 2020). In aninteresting study, Li and his group showed that increase in self-mediated

pH facilitates the biosorption of cadmium by *Bacillus cereus* Cd01 (Li et al. 2018). Using three different sulphate reducing bacteria, most promising recovery of platinum and palladium were obtained at pH 3 (De Vargas et al. 2004). Some microbes, such as *Bacillus amyloliquefaciens* and *Oceanobacillus profundus*, have demonstrated optimal biosorption at neutral to slightly alkaline pH (Wandira et al. 2020; Patel and Chandel 2025), highlighting the diversity in microbial response and the importance of tailoring pH conditions to specific biosorbents. The biosorption efficiency of *B. cereus* F4810/72 was highest (~79%) at 50 °C, but dropped sharply on further increasing the temperature (Fig. 5). Studies involving different microorganisms such as *Bacillus licheniformis*, *Lactobacillus brevis*, and *Pseudomonas monteii* reported optimal biosorption between 30 and 45 °C (Dai et al. 2019; Patel and Chandel 2015; Canaza et al. 2021). In a study by Ozdemir and his group, the temperature obtained for maximal biosorption was observed near the growth temperature of the bacteria used in the study (Ozdemir et al. 2013b). The influence of temperature on Cu(II) biosorption was examined using two distinct bacterial strains. It was observed that *Arthrobacter* sp. Sphe3 exhibited a decline in metal uptake as the temperature increased from 20 to 40 °C, whereas *Bacillus sphaericus* showed an enhanced uptake capacity over the same temperature range (Aryal et al. 2012). Additionally, the highest sorption efficiency for Cr(III) by *Bacillus subtilis* (Aravindhan et al. 2012) and for Mn(II) by *Anoxybacillus amylolyticus* (Ozdemir et al. 2013a) was recorded at a comparatively elevated temperature of 60 °C. This observed improvement in biosorption performance up to 50 °C is likely due to increased kinetic energy of the system, which facilitates more effective collisions between the metal ions and the active binding sites on the bacterial biomass. Additionally, higher temperatures can enhance the flexibility of cell wall polymers, thereby exposing more functional groups responsible for metal binding. However, a further rise in temperature beyond 50 °C led to a marked decline in biosorption efficiency due to denaturation or structural alteration of key functional groups involved in metal binding, or the destabilization of the bacterial surface architecture. Excessive thermal energy may also disrupt the physical integrity of the biosorbent or alter metal ion speciation, both of which can negatively affect biosorption performance. However, most of the scientific studies determined the optimum temperature for maximum removal of heavy metals at a temperature ranging between 25 °C and 35 °C using diversified bacterial strains (Jencarova and Luptakova 2012; Rani et al. 2010; Hussein et al. 2004; Liu et al. 2004; Karakagh et al. 2012; Sulaymon et al. 2013). Agitation speed also affects biosorption by influencing mass transfer rates. Maximum biosorption (79%) was observed at 80 rpm (Fig. 6). Enhanced mixing increases contact between biomass and metal ions, though overly high speeds may cause shear damage or prevent sufficient binding due to turbulent mixing. Interestingly, this optimal speed is consistent with the agiatation speed reported in other studies, such as biosorption of cadmium using *Klebsiella* sp. (Hou et al. 2015), suggesting biosorbent-specific responses. In another study, *Streptomyces rimosus* biomass was used to remove nickel at an optimum stirring condition of 250 rpm (Selatnia et al. 2004). Tarangini and Satpathy have investigated that nearly 85% chromium was removed using attenuated cultures of *Pseudomonas aeruginosa* and *Bacillus subtilis* at an agitation speed of 300 rpm (Korumilli and Satpathy 2009). Controlled agitation is vital for maintaining high biosorption efficiency in treatment systems. Under optimized conditions—250 µg/mL Cr concentration, 4 g/L biomass dose, pH 3.0, 50 °C, 70 min contact time, and 80 rpm agitation *Bacillus cereus* F4810/72 achieved a maximum biosorption efficiency of nearly 79%. In comparison to advanced composite adsorbents, such as ligand-based materials and chitosan-coated composites, which offer high adsorption capacities (155–195 mg/g) (Sheikh et al. 2023; Kubra et al. 2023; Awual et al. 2015; Awual et al. 2016; Rasee et al. 2023; Hasan et al. 2023), the biomass-based adsorbent in this study demonstrates moderate capacity. However, it excels in terms of cost, reusability, environmental safety, and biodegradability. These advantages make it a viable option for large-scale applications, especially where resource limitations exist. Adsorption isotherm results demonstrated that the biosorption process follows monolayer adsorption on a homogeneous surface, as supported by the superior fit of the nonlinear Langmuir model (Table 1). The discrepancy between linearized and nonlinear Langmuir Q_{max} values highlights the tendency of linearization to overestimate adsorption capacity. The Freundlich model suggested favorable adsorption ($n > 1$), reflecting heterogeneity in adsorption sites,

but higher SSE, RMSE, and χ^2 values indicate that this model is less suitable for describing the system. This analysis demonstrates that error function metrics (SSE, RMSE, ARE, χ^2) provide more reliable guidance for model selection than R^2 alone. Nonlinear fitting is recommended for accurate estimation of adsorption parameters. The kinetic data describing chromium uptake by the biomass were evaluated using both pseudo-first-order (PFO) and pseudo-second-order (PSO) models. The PSO model exhibited the best fit to the experimental data, with a higher regression coefficient ($R^2 = 0.9956$) compared to the PFO model ($R^2 = 0.9872$). The qe value predicted by the PSO model (15.71 mg/g) also closely matched the experimentally observed equilibrium uptake, indicating its superior predictive accuracy (Supplementary Information 1). These results suggest that chromium biosorption onto the biomass is predominantly governed by a chemisorption-controlled mechanism involving valence forces or electron sharing between functional groups and chromium ions. The good agreement of the PSO model with the kinetic data further supports the rapid occupation of available active sites followed by a slower approach to equilibrium. This supports the suitability of *Bacillus cereus* F4810/72 for complex effluent treatment. Effective regeneration of biosorbents is critical for economic feasibility. The study found that 2 M HCl eluted ~ 80% of adsorbed chromium, outperforming 0.5 M and 1 M NaOH and HCl (Fig. 7). In contrast, our previous work demonstrated that 1 M HCl was highly effective in the elution of chromium. Further elution experiments revealed that a mixture of 0.2 M HNO_3 and Ascorbic acid was more effective as a promising eluant than EDTA and other acids (Fig. 8). Various studies reported the use of Ascorbic acid as a promising eluant for the removal of metal ions from contaminated environment (Thinh et al. 2021; Hokkanen et al. 2013; Li et al. 2020). A study also reported the use of an immobilized plant biomass for metal removal (Kamala et al. 2005), suggesting the reusability and long-term applicability of the bacterial biosorbent. Also, the regeneration study demonstrated that bacterial biomass beads maintained consistent chromium ion elution over three consecutive cycles, exhibiting greater stability compared to raw biomass (Table 2 and 3) which is consistent with our previous findings (Bhattacharjee et al. 2023). These findings indicate the potential reusability of the immobilized bacterial biomass foreffectiveremoval of metal ions from contaminated aqueous solutions. Overall, the findings underscore the potential of *Bacillus cereus* F4810/72 as an efficient, low-cost, and eco-friendly biosorbent for chromium removal from industrial effluents. While further enhancement and large-scale validation are needed, the biosorbent offers a practical.

Table 2 Regeneration of raw biomass

Number of cycles	Uptake of chromium (mg/g)	Elution of chromium (mg/g)
1	91.4 ± 3.6	87.5 ± 4.2
2	86.4 ± 1.9	80.2 ± 3.7
3	76.6 ± 2.8	71.2 ± 2.6

Table 3 Reusability of bacterial biomass

Number of cycles	Uptake of chromium (mg/g)	Elution of chromium (mg/g)
1	98.8 ± 4.6	98.3 ± 3.8
2	97.6 ± 3.8	97.4 ± 3.2
3	96.9 ± 3.2	96.4 ± 4.1

Conclusion

This study demonstrates that *Bacillus cereus* F4810/72 is a highly effective and reusable biosorbent for the removal of hexavalent chromium from tannery effluents. Systematic optimization of operational parameters revealed that the strain achieved a maximum Cr(VI) removal efficiency of 79–80% under the optimized conditions of 250 µg/mL initial metal concentration, 4 g/L biomass dosage, pH 3, 50 °C, 70 min contact time, and 80 rpm agitation. Initial concentration studies showed a clear rise in biosorption from 19.46% at 10 µg/mL to 79.8% at 250 µg/mL, with subsequent decline at higher concentrations due to site saturation. Contact time experiments confirmed equilibrium at 70 min, while biomass optimization showed improved uptake up to 3–4 g/L, beyond which no significant enhancement occurred. pH-dependent analysis identified

pH 3 as optimal, yielding the highest adsorption (~80%), and temperature profiling revealed that biosorption peaked at 50 °C with ~78% uptake. Adsorption isotherm modeling provided further mechanistic insights. The nonlinear Langmuir model, which best described the adsorption behavior, indicated a maximum monolayer adsorption capacity (Q_{max}) of 28.57 mg/g, while the Freundlich isotherm ($R^2 = 0.98$) suggested heterogeneous multilayer adsorption with favorable intensity (1/n = 0.598). These findings confirm the presence of diverse functional groups on the bacterial biomass contributing to strong binding and complex formation with Cr(VI) ions. Regeneration studies underscore the economic viability of the biosorbent. 2 M HCl achieved ~80% desorption, outperforming weaker eluents, while a 0.2 M HNO_3–ascorbic acid mixture yielded the highest elution efficiency among all tested agents. Immobilized biomass beads exhibited remarkable reusability, retaining > 96% uptake across three cycles, significantly outperforming raw biomass, which showed moderate decline (from 91.4 mg/g to 76.6 mg/g over three cycles). Overall, the study highlights *B. cereus* F4810/72 as a cost-effective, efficient, and reusable biosorbent capable of substantial Cr(VI) removal and recovery. The strain's high adsorption capacity, rapid equilibrium time, strong performance under acidic conditions and excellent regeneration potential make it a promising candidate for scalable, environmentally sustainable remediation of chromium-contaminated industrial effluents. The quantitative outcomes of this research provide a foundation for pilot-scale applications and future developments in bioremediation strategies for heavy metal pollution.

Acknowledgements The authors express their deep gratitude of thanks to the Department of Microbiology, Kingston College of Science; Department of Microbiology, St. Xavier's College, Kolkata; and Department of Biotechnology, National Institute of Technology Durgapur. The authors are also deeply indebted to Department of Microbiology, West Bengal State University, Rev. Dr. Dominic Savio, S.J., Principal and Rector, St. Xavier's College (Autonomous), Kolkata, Mrs Uma Bhattacharjee, Secretary, Kingston Educational Institute, Mr. Tipam Bhattacharjee, President, Kingston Educational Institute for their constant support and guidance throughout.

Author contributions The experimental design was performed collectively by all the authors. The preparation of reagents and samples was done by AB, DC, TS, SK, SR, KD, KM. Data collection and analysis, was done by AB, SK, SP, TS. The manuscript was written by AB and SRC with inputs from all the authors. AB: draft manuscript preparation, methodology, data collection, and editing; DC: Methodology; SP: Data Analysis; TS: Conceptualization and overall supervision; SK: Sample Preparation, Data Analysis and overall supervision; AKM: Overall supervision and resources; SRC: Manuscript preparation; TB: Manuscript editing; SR: Methodology and Data collection; KD: manuscript editing; All the authors read and approved the final manuscript.

Declarations

Conflict of interest The authors declare no competing interests.

Consent for publication All the authors consent to the consideration for this manuscript for publication.

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